

Full Looped Transformer

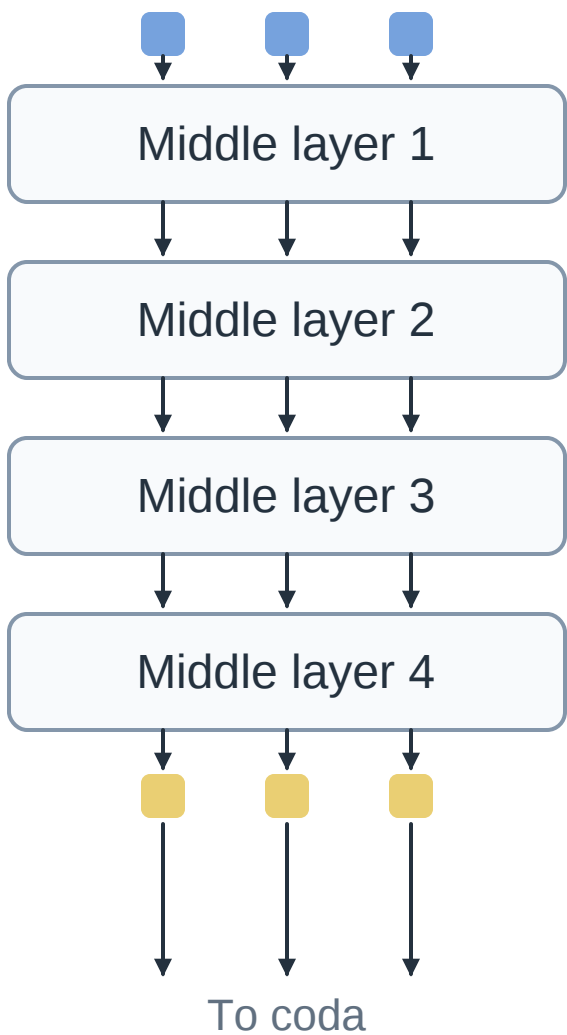
What crosses the loop boundary?

Input tokens → Embedding E → RMSNorm → Prelude (2 layers)

Initial state Hidden state Feedback

(a) Dense Transformer

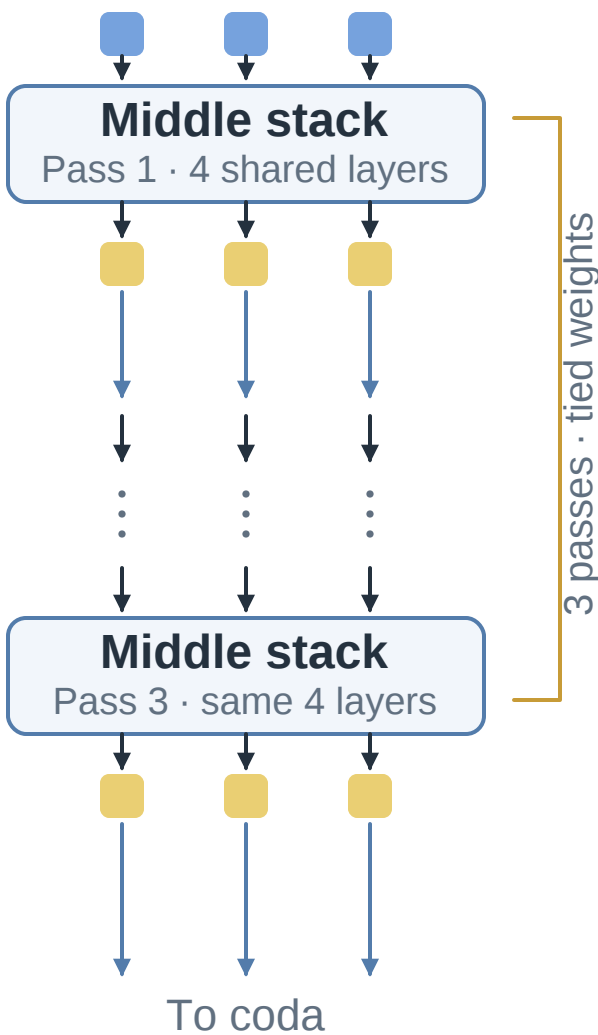
8 physical / 8 unrolled layers



Val. loss ↓ Zero-shot ↑
2.9064 **42.53**
One visit per layer

(b) Plain Loop

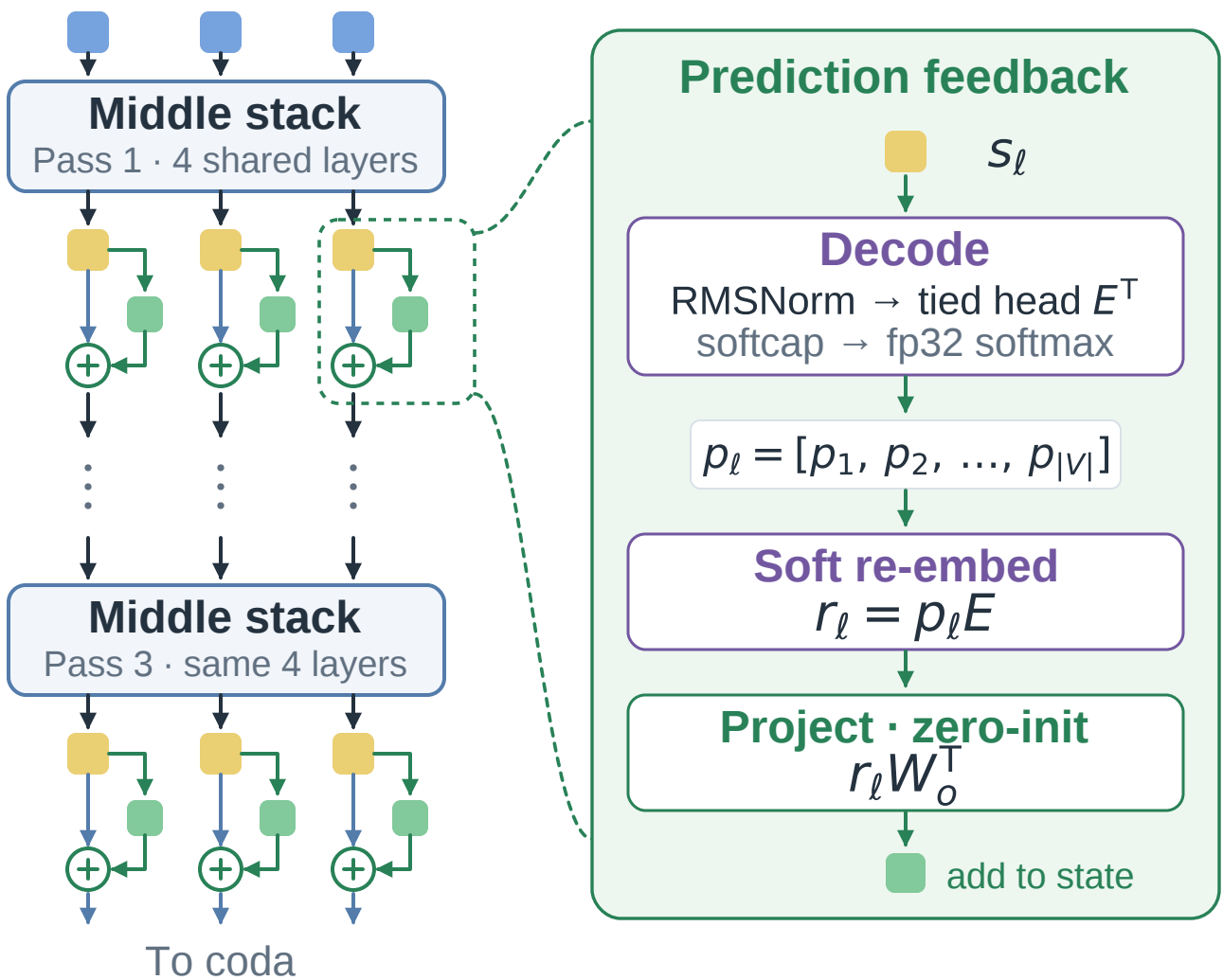
8 physical / 16 unrolled layers



Val. loss ↓ Zero-shot ↑
2.8647 **43.77**
Hidden-state carry

(c) Full Loop (ours)

8 physical / 16 unrolled layers



Val. loss ↓ Zero-shot ↑
2.8145 **44.51**
-0.0502 loss / +0.74 pp vs. plain

Output Coda (2 layers) → RMSNorm → tied head E^T → softcap → softmax → p_{out}

Looped models: pass 2 omitted; 3 passes total. Full loop adds feedback after all 3.

10B tokens · 6-task mean (%)